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BERT and RoBERTa Models for Enhanced Detection of Depression in Social Media Text

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Abstract

Depression is a disease that affects everyone, both young and old. This mental illness not only affects the surrounding environment but everyone. Depression is characterized by deep sadness, behavioral changes and many other actions that are risky for people. In this research we try to solve the problem of detecting depression using Natural Language Processing (NLP) approaches, these two methods are Bidirectional Encoder Representations from Transformers (BERT) and Robustly Optimized BERT Approach (RoBERTa), where these two methods are used to detect posts made in reddit. The dataset is taken from Kaggle. The results obtained found that the average use of BERT and RoBERTa resulted in a high accuracy value of around 98% and with a well balanced precision, recall and F1-Score ratio. This research shows that there is a possibility of using BERT and RoBERTa in depression detection.

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1. Introduction

Depression is a significant global mental health problem, impacting a person's emotions, behavior, and daily activities. This mood disorder can cause feelings of sadness, anger, and disinterest in previously enjoyed activities, affecting people of all ages and backgrounds. According to the World Health Organization (WHO), more than 260 million people worldwide suffer from depression, making it a leading cause of disability and a major contributor to the overall

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global burden of disease (WHO2023). The condition not only reduces quality of life but also increases the risk of various physical health problems [1].

Mental health issues, such as depression, are highly prevalent in Indonesia. According to survey that conducted by Dokter Spesialis Kedokteran Jiwa Indonesia (PDSKJI) and Badan Penelitian dan Pengembangan Kesehatan (Litbangkes) indicates that a significant portion of the population is currently grappling with anxiety disorders and depression. These mental health conditions are further intensified by social and environmental stressors. These findings emphasize the immediate requirement for comprehensive mental health programs and interventions in the country. [2].

Timely identification of depression has numerous advantages, greatly enhancing the likelihood of successful therapy and recuperation. Early detection can help in avoiding more severe conditions and help people with depression to get first aid as soon as possible. Furthermore, proper treatment can help sufferers to live a better life.

Various techniques have been used to identify depression, including assessing clinical data or electronic health records [3, 4, 5], observing personal disclosures on social media [6, 7, 8], and observing individuals' speech patterns [9, 10]. This research aims to investigate social media data, as more and more users are using social media platforms as a means to express their emotions. As a result, social media platforms such as Twitter or Reddit have been determined as appropriate sources for obtaining related data.

The aim of this study is to perform a comparative examination of two transformer-based methods to obtain a thorough comprehension of the contextual factors associated with mental health challenges as expressed on social media sites.

This article is organized into five different sections. Section I serves as an Introduction, Section II delves into Previous Research, Section III outlines the Research Methodology, Section IV provides the Experiments and Results, and Section V provides the research conclusions.

2. Previous Research

Zaman et al. [11] discuss the worldwide occurrence of depression, highlighting the dangers linked to severe instances and the significance of timely identification and intervention. The study examines the role of social media platforms in detecting initial manifestations of depression through user posts. The research endeavors to utilize a highly optimized RoBERTa model to forecast different degrees of depression (mild, moderate, and severe) by specifically analyzing Twitter data. The algorithm exhibits a 90% accuracy rate in identifying multilayer depression, highlighting the possibility of using social media to aid in mental diagnosis and suicide prevention.

Suhas et al. [12] used Text Analysis approach. They implemented various machine learning algorithms for depression detection. The result concluded that the Random Forest Classifier performs better compared to other algorithms. On the other hand, Kanaan et al. [7] developing an application that utilizes a neural network trained on social media data to accurately forecast depression and suicidal tendencies in real-time. The study conducted experiments using six deep learning models, which incorporated approaches using word embedding. The results indicate that the CNN+LSTM model, which includes word embeddings, performed better than other models. It achieved an accuracy rate of 97.56% after 15 epochs, closely followed by an LSTM model with 97.48% accuracy.

Fatimah et al. [2] utilized classification algorithms, namely Naïve Bayes, SVM, Logistic Regression, and Random Forest, to identify signs of psychological discomfort. The research identified 612 tweets that indicated psychiatric difficulties, with 168 messages particularly indicating anxiety concerns. Two models were created to categorize these concerns: Model 1, employing the Naïve Bayes method, exhibited notable efficacy with an accuracy of 74.36%, but Model 2, utilizing the SVM algorithm, demonstrated a slightly superior accuracy of 76.42%. These findings highlight the capacity of social media as a means of diagnosing and tackling mental health concerns, such as depression, in times of crisis like the Covid-19 outbreak.

Chowanda et al. [13] emphasizes the limited availability of datasets and studies that specifically examine non-English languages, such as Indonesian. This is crucial since these languages exhibit distinct ways of expressing ideas and emotions, which are influenced by cultural, racial, and religious factors. The study focuses at seven different models and finds that putting Bidirectional Long Short-Term Memory (Bi-LSTM) and the Bidirectional Encoder Representations from Transformers (BERT) architecture makes the models much better at finding signs of depression.

Karna et al [14]. looks into the idea of using machine learning to find signs of depression in social media posts. The study describe a way to use transformer-based models (BERT and RoBERTa) along with common deep learning designs like LSTM, CNN with BiLSTM, and RNN. Before comparing different model setups, the study goes through a thorough process of getting the data ready and embedding words. The results show that mixing BERT and Dense layers, as well as RoBERTa and Dense layers, works much better than other methods, with an amazing 98 percent success rate.

In the context of the increasing relevance of machine learning and deep learning algorithms highlighted by previous research, this study aims to implement Bidirectional Encoder Representations from Transformers (BERT) and RoBERTa models, as conducted by [14] and Zaman et al. [11], into our dataset. These implementations are intended to explore the efficacy of advanced NLP techniques in analyzing and understanding the dataset more effectively, in line with the innovative approaches established by the referenced studies.

3. Methodology

In this section, we explained the methodology including the dataset, data pre-processing, BERT and RoBERTa.

3.1. Dataset

The dataset has two features: 'clean_text', and 'label', which includes the class either depression ('1') or not ('0'). The total of the entries are 1203 entries.

The description suggests that Fig 1 represents the distribution of classes within the dataset.

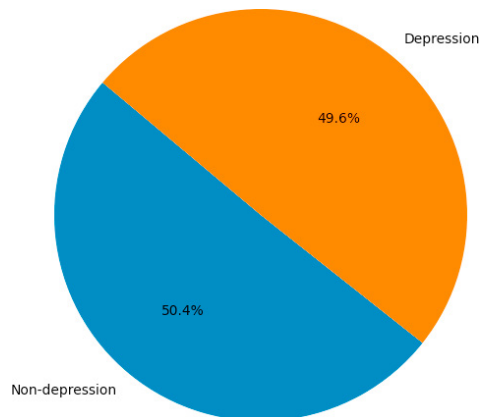


Fig. 1: Data Distribution of Depression

The dataset is evenly divided into two classes: 'Non-depression' and 'Depression'. The 'Non-depression' class represents 50.4% of the dataset, while the 'Depression' class makes up 49.6%.

3.2. Pre-processing

Tokenization is done during the preprocessing stage. We use a BERT or RoBERTa Tokenizer in this step to change the text into a style that works with the BERT Model. After the tokenization process is done, there is a list of number tokens that for each word. An attention mask is also made to tell the difference between real tokens and padding tokens. This makes the model understand the length of sequence and data for each input .

3.3. Bidirectional Encoder Representations from Transformers (BERT)

Transformer, an attention mechanism that discovers contextual relationships between words (or sub-words) in a text, is utilized by BERT. Transformer, in its rudimentary state, comprises two distinct mechanisms: an encoder responsible for processing the input text, and a decoder designed to generate a task prediction. In order to produce a language model, BERT requires solely the encoder mechanism.

In contrast to directional models, which operate by sequentially reading the input text from left to right or right to left, the Transformer encoder scans each word in its entirety at once. As a result, it is classified as bidirectional, although a more precise description would be that it is unidirectional. The aforementioned attribute enables the model to acquire knowledge regarding the context of a word by analyzing its immediate and peripheral environment.

The input consists of a series of tokens that are initially vectorized before undergoing neural network processing. Each vector in the sequence of output vectors of size H represents an input token with the corresponding index.

BERT uses two training strategies. The initial stage involves the Masked Language Model. A [MASK] token is used as a replacement for 15% of the words in each sequence before feeding them into BERT. Afterwards, the model aims to predict the starting value of the hidden words by using the context obtained from the visible words in the sequence. The prediction of the output words technically entails the following: superimposing a classification layer onto the encoder output; By converting the output vectors to the vocabulary dimension through multiplication with the embedding matrix, Using softmax to compute the probability of each word in the vocabulary. The predicted value of non-masked words is disregarded by the BERT loss function, which only considers the prediction of masked values. Consequently, the model converges at a sluggish rate compared to directional models; however, this drawback is counterbalanced by its heightened awareness of the context[15].

The second phase in the BERT training technique is Next Sentence Prediction. The model is given pairs of sentences and asked to determine if the second sentence logically follows the first sentence in the document. A randomly selected text from the corpus is used as the second sentence in the remaining 50% of input pairings during training. Half of the input pairings had the second sentence as the document's next sentence. It is presumed that the randomly selected sentence is unrelated to the original. The input is processed before being supplied to the model to help it distinguish between the two sentences during training. [15]:

- A [SEP] token is appended to the conclusion of each sentence, and a [CLS] token is appended to the start of the initial sentence.
- Each token is endowed with a sentence embedding denoting either Sentence A or Sentence B.
- Each token is assigned a positional embedding that specifies its location within the sequence.

To determine the connection between the first and second sentences, the following techniques are necessary [15]:

- The input sequence is processed in its entirety by the Transformer model.
- The result of the [CLS] token is converted into a vector with the shape 2×1 by employing a basic classification layer comprised of learned matrices representing weights and biases.
- Employing softmax to compute the probability of IsNextSequence.

The BERT model trains Masked LM and Next Sentence Prediction simultaneously to minimize the combined loss function of both techniques. The BERT architecture is illustrated in Fig 2.

BERT is available in two distinct model sizes: BERT Base and BERT Large.

Table 1: Comparison between BERT Base and BERT Large

	BERT Base	BERT Large
Layers	24	112
Hidden Size	768	1024
Heads	12	16
Parameters	110M	340M

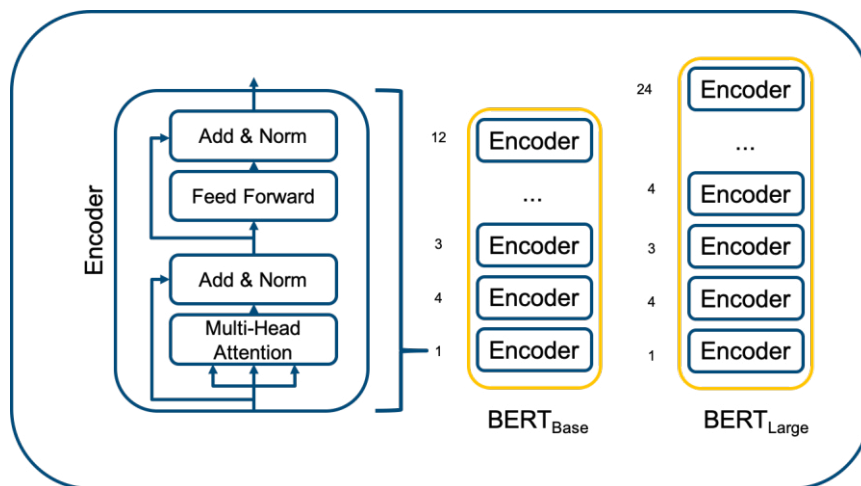


Fig. 2: BERT architecture

3.4. Robustly Optimized BERT Approach (RoBERTa)

RoBERTa, although it shares a similar structure with BERT, incorporates various alterations with the goal of improving the model's performance. An important change has been made by removing the Next Sentence Prediction (NSP) aim. The initial BERT model employed the Next Sentence Prediction (NSP) task during training to determine if two text segments were from the same or separate texts. Nevertheless, the creators of RoBERTa discovered that removing the NSP loss either equals or slightly enhances the performance on subsequent tasks. Furthermore, RoBERTa underwent training with greater batch sizes and longer sequence lengths compared to BERT. Specifically, it was trained for 125 steps using 2,000 sequences and for 31,000 steps with batches consisting of 8,000 sequences. This strategy not only increased the confusion on the masked language modeling objective, but also improved the accuracy of end tasks. The larger batch size enabled more effective parallelization through dispersed training. RoBERTa introduces a significant shift in its approach to masking compared to BERT. Instead of using a fixed mask during data preprocessing, RoBERTa utilizes dynamic masking. This includes duplicating the training data and applying different masks to it 10 times over the course of 40 epochs. This modification guarantees that the model does not depend on a solitary, unchanging mask, hence improving its ability to learn and generalize [16].

4. Experiment and Result

4.1. Experiment Setting

In this study, we utilized Google Colab for conducting our research. The dataset employed was sourced from <https://www.kaggle.com/datasets/infamouscoder/depression-reddit-cleaned>. The table below illustrates the hyperparameters used in this current study.

Table 2: Hyperparameters used in depression detection

Hyperparameter	Value
Learning Rate	2×10^{-5}
Number of Epochs	5
Dropout Rate	0.1
Sequence Length	128
Optimizer	AdamW

4.2. Experiment Result

Table 1 will display the results from the experiments conducted with the two methods used.

Table 3: Result of Depression Detection

	Accuracy	Precision	Recall	F1
BERT	98	0.98	0.98	0.98
RoBERTa	98	0.99	0.97	0.98

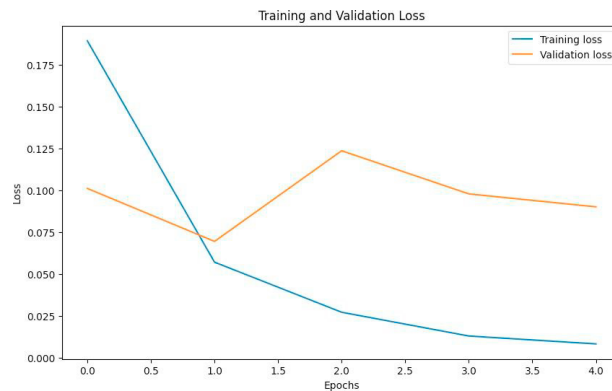


Fig. 3: graphic loss of BERT

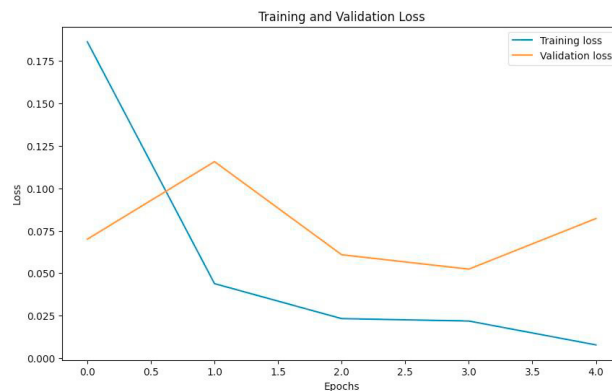


Fig. 4: graphic loss of RoBERTa

According to Figure 3, the training loss is initially higher than the validation loss in the first epoch. However, the training loss gradually decreases in subsequent epochs, while the validation loss increases in the second epoch. The occurrence of this pattern of divergence usually indicates overfitting, which is a situation where the model performs poorly on validation data that it has not been trained on. However, it is not conclusive to claim the existence of overfitting based on the results of this experiment, given the evaluation was conducted for only five epochs.

Similarly, the results obtained from the RoBERTa model, as shown in Figure 4, demonstrate a same pattern. However, upon closer examination, it is evident that there is a smaller difference between the training and validation losses compared to what was observed with BERT. The narrower gap indicates that RoBERTa has superior resilience compared to BERT.

5. Conclusion

Depression has a substantial impact on mental health worldwide, affecting people's feelings, actions, and daily activities. This mood illness can elicit intense feelings of melancholy, irritation, and lack of interest in things that were previously loved, impacting persons from all demographic groups. For this study, we utilized a dataset sourced from Kaggle that pertains to depression on the Reddit platform. Our investigation involved the application of two techniques, namely BERT and RoBERTa. The experimental results indicate that both proposed strategies yield favorable outcomes for the depression dataset, with each method attaining an accuracy rate of 98%. Furthermore, both techniques demonstrated equitable performance indicators, including precision, recall, and F1-Score. In future research, scientists intend to employ similar approaches on a wider range of datasets and include the Indonesian language.

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